

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
ACADEMIC YEAR 2021/2022
JUNE EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

TUESDAY, 28 JUNE 2022

TIME: 9.00 AM – 11.00 AM (2 HOURS)

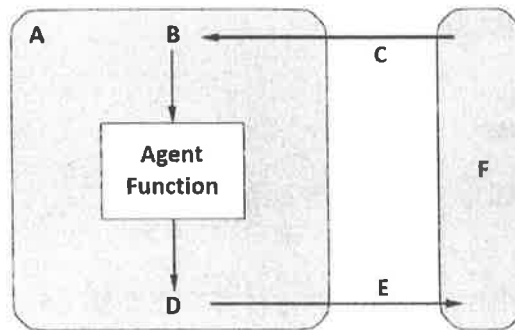
DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

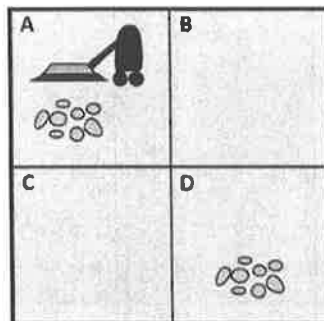
Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) The word *agent* comes from the present participle *agentis* (i.e., *the one doing*) of the Latin verb *agree*, which carries the meaning of *do* or *make*. However, in the field of Artificial Intelligence (A.I.), it is defined differently.
- (i) Define the term *rational agent* in the context of A.I. (3 marks)
- (ii) The idea of an *agent* can also be illustrated as in **Figure 1**. Name the **SIX (6)** missing labels indicated by letter A, B, C, D, E, F and G. (6 marks)

**Figure 1. An illustration of agent**

- b) Jack plans to create a rational vacuum agent that will help him clean up all of the dirty squares. His initial idea is to place the agent in an empty room consists of 4-squares only at the early stage of the development, as depicted in **Figure 2**.

**Figure 2. A 4-square vacuum world**

However, before Jack can start designing the agent, he needs to formulate the problem first, so that he can have a better understanding of the problem. From there, he will also be able to visualize the state space of the problem, which is the set of all states reachable from the initial state by any sequence actions.

- (i) Define *agent function*. Then, give an example of the *agent function* for the vacuum agent. (2 + 2 marks)

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1 b) (Continued)****Table 1. Problem Formulation**

Initial state	Any state can be designated as the initial state.
Actions	Each state has just 3 actions: Left, Right, and Suck.
Transition model	The actions have their expected effects, except that moving Left in the leftmost square, moving Right in the rightmost square, and Sucking in a clean square have no effect.
Goal test	This checks whether the current square is clean.
Path cost	Each step costs 1, so the step cost is the number of steps in the path.

- (ii) A problem formulation shown in **Table 1** consists of three incorrect descriptions. Identify which **THREE (3)** and correct them by rewriting them. (3 + 3 marks)
- (iii) A state space is the resultant of the combination of three elements. Identify these **THREE (3)** elements. Then, calculate the total number of possible states for the problem specified in **Question 1 b)**. (3 + 3 marks)

[Total: 25 marks]

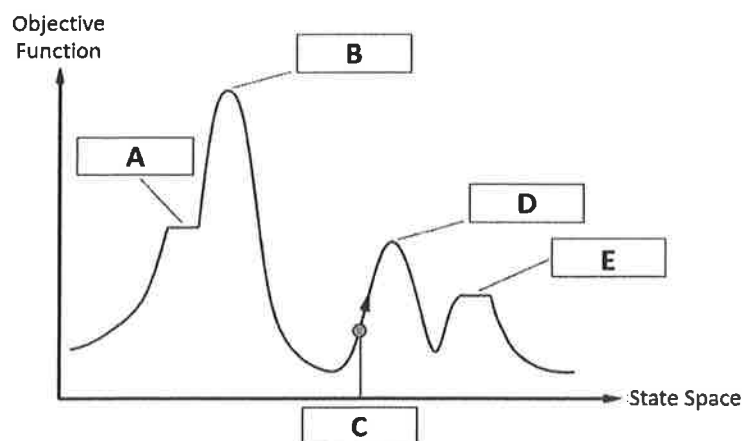
Question 2

Hill-climbing search is an example of local search strategies. The search continually moves in the direction of increasing value, and terminates when it reaches a peak where no neighbour has a higher value. Thus, it can be implemented with a very simple algorithm as provided in **Figure 3**.

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function Hill-Climbing(problem) returns a state that is a local maximum
    neighbour <-- a highest-valued successor of current
    current <-- neighbor
    loop do
        current <-- MAKE-NODE(problem.INITIAL-STATE)
        if neighbour.VALUE ≤ current.VALUE then return current.STATE

```

Figure 3. Hill climbing algorithm**Figure 4. State-Space Landscape**

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 2 (Continued)**

To understand what may happen to hill-climbing during the search, one can refer to a state-space landscape illustrated in **Figure 4**, which is a graphical representation of the algorithm.

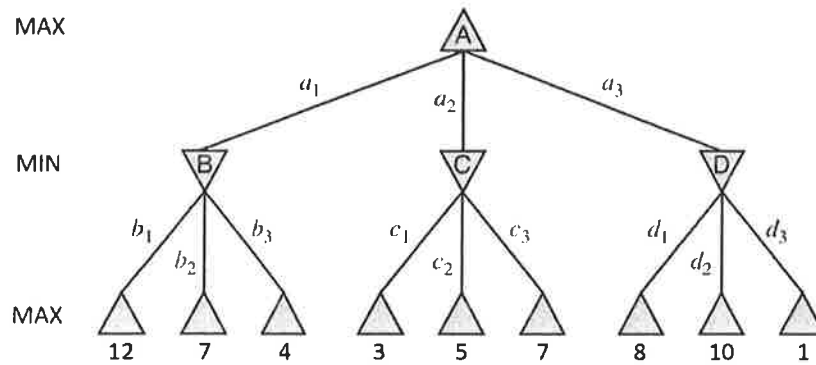
- a) Unfortunately, the hill-climbing algorithm provided in **Figure 3** does not work properly because the five lines of the algorithm (contained inside the body) are arranged in the wrong order. Put them in the right order with the use of an appropriate indentation so that the algorithm can function properly. (5 marks)
- b) The state-space landscape provided in **Figure 4** has some missing labels. Name the **FIVE (5)** missing labels. Then, briefly describe each of them. (5 + 5 marks)
- c) Discuss how a hill-climbing strategy can outperform most of the classical search strategies in **FOUR (4)** ways. (4 marks)
- d) Despite its strengths, hill climbing search often gets stuck at a non-optimal solution. Identify **THREE (3)** locations in state-space landscape illustrated in **Figure 4** where this can happen. (3 marks)
- e) Many variants of the hill climbing strategy have been proposed to address its limitation. **THREE (3)** of them are *stochastic hill climbing*, *first-choice hill climbing* and *randomly-restart hill climbing*. Briefly describe how they differ from hill climbing strategy. (3 marks)

[Total: 25 marks]

Question 3

Michelle is developing a rational agent that is capable of playing chess game (without clock). She has just completed the *PEAS* description for her chess agent. Now, she is going to work on the properties of the environment in which the agent will be operating. Once this is done, she will start looking into the *Minimax algorithm*, which she believes will enable her agent to calculate the best move during the game.

- a) *PEAS* is an acronym that stands for four elements, name the **FOUR (4)** elements. (4 marks)
- b) In general, environments have already been categorized into six dimensions. Identify **THREE (3)** types of environments the chess agent will be situated. Then, provide an explanation for each of them. (3 + 6 marks)
- c) Explain why the two players in a chess game are referred to as *MAX* and *MIN*. (4 marks)
- d) Given a game tree in **Figure 5** below, compute the minimax value for node B, C and D, and provide a brief explanation for each of them. Then, identify *MAX*'s best move at the root node followed by a brief explanation as well. (8 marks)

Question 3 d) (Continued)**Figure 5. A game tree**

[Total: 25 marks]

Question 4

Problem solving agents seem to know something but they cannot go beyond that. Humans, on the other hand, tend to make use of reasoning to enable them to do a lot more things on top of what they already knew. In A.I., this approach to intelligence is embodied in knowledge-based agents.

- Describe **TWO (2)** capabilities of a knowledge-based agents which are absent in problem-solving agents. (2 marks)
- Knowledge-based agents are composed of two central components: a *knowledge base* and an *inference system*. Describe the meaning of these two components in a way that they are related to each other. (5 marks)
- Propositional logic is the simplest form of logic. In propositional logic, five *logical connectives* are commonly in use such as $\neg$, $\wedge$, $\vee$, $\Rightarrow$ and $\Leftrightarrow$. Examples provided below show the usage of these three symbols.

$$S_1: (\text{Pit}_{1,2} \vee \text{Pit}_{2,1}) \Rightarrow \text{Breeze}_{1,1}$$

$$S_2: \text{Monster}_{1,3} \Leftrightarrow \neg \text{Monster}_{2,2}$$

- Translate the two sentences S_1 and S_2 into English sentences. (3 marks)
- Write down a symbol that represents *logical equivalence* and identify a requirement to be satisfied to make two sentences α and β logically equivalent to each other. Then, demonstrate $\neg(P \wedge Q)$ is logically equivalent to $\neg P \vee \neg Q$ with the use of truth table. (1 + 1 + 5 marks)
- A logical inference procedure can be implemented via two different approaches: *model checking* and *theorem proving*. Explain why one approach is better than another one in terms of efficiency. (5 marks)

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4 c) (Continued)**

- (iv) Theorem proving is supported by three important concepts. Name the **THREE (3)** concepts.
(3 marks)

[Total: 25 marks]